MIL-I-23479B

16 October 1967

Superseding
MIL-I-23479A(WP)
31 March 1965

MILITARY SPECIFICATION

INDICATOR, PRESSURE, 1 1/2-INCH, INTEGRALLY LIGHTED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers design requirements and all performance requirements for the procurement of single and dual hermetically sealed, integrally lighted, remote indicating, pressure indicators.

1.2 Classification - The indicators shall be furnished in the following ranges as specified in MS17996.

Range	Single Indicator 1 1/2 inch size	Dual Indicator 1 1/2 inch size
0-5000 Hyd	MS17996-1	MS17996-5
0-100 Oil	MS17996-2	MS17996-6
0-2500 Hyd	MS17996-3	MS17996-7
0-300 Oil	MS17996-4	MS17996-8
0-50 Boost	MS17996-9	MS17996-10
0-150 Oil	MS17996-11	MS17996-14
0-200 Oil	MS17996-12	MS17996-15
0-250 Oil	MS17996-13	MS17996-16

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

DD-G-451	Glass, Flat and Corrugated, for Glazing, Mirrors, and other uses
PPP-B-601	Boxes, Wood, Cleated-Plywood

FSC 6620

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SPECIFICATIONS

Federal (Continued)

PPP-B-636	Boxes, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-I-7057	Indicators, Synchro, Aircraft, General Specification for
MIL-T-7748	Transmitters, Pressure, Synchro, Aircraft, 320 - Degree Movement, High Temperature, General Purpose
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-C-26482	Connectors, Electric, Circular, Miniature Quick Disconnect

STANDARDS

Federal

FED STD NO. 595	Colors

Military

MIL-STD-130	Identification Marking of U S Military Property
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods
MIL-STD-781	Reliability Test Exponential Distribution
MS 17996	Indicator, Pressure, 1 1/2 inch, Integrally Lighted

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Qualification - The indicator furnished under this specification shall be a product which has been tested and meets the qualification inspections specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Order of precedence - The requirements of MIL-I-7057 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.2.1 Where the general specification and this specification conflict, this specification shall govern.

3.3 Design and construction - The indicator shall be designed to provide remote indication of pressure in pounds per square inch when connected to a transmitter conforming to MIL-T-7748. The calibration of pressure versus angular degrees shall be linear within ±0° 15' of arc. The indicator shall be integrally lighted as specified in MIL-I-7057 and MIL-L-25467.

3.4 Dial - The dial shall be marked as shown on figure 1. Unless otherwise specified all visible portions of the dial shall be lusterless black Color No. 37038 of FED. STD. NO. 595 and all markings on the dial shall be lusterless white Color No. 37875 of FED. STD. NO. 595.

3.5 Pointer(s) - The pointer shall be as specified on MS17996. The pointer shall be suitable for integral lighting. The shaded portion of the pointer shall be finished in a lusterless white conforming to Color No. 37875 of FED. STD. No. 595.

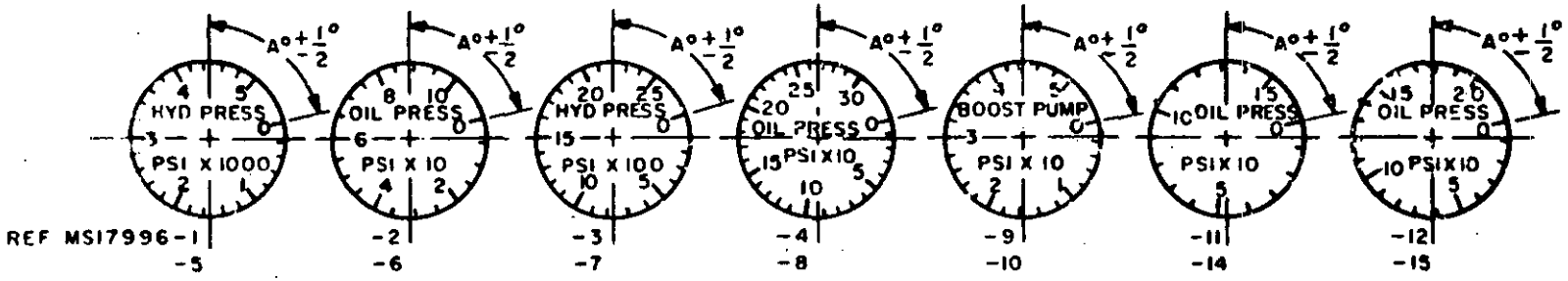
3.6 Performance - The indicator shall perform satisfactory when subjected to the tests specified in Section 4.

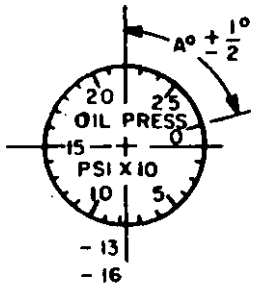
3.6.1 Reliability in mean-time-between-failures (MTBF) - The indicator shall have 1000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.7 Case - The outline dimensions of the indicator case shall conform to MS17996. The case shall be hermetically sealed and the design shall be such that the mechanism may be removed from the case, replaced, and the case resealed without the use of special tools and fixtures.

3.7.1 Cover glass - The quality of the cover glass shall be in accordance with DD-G-451, Type II, Quality AA. Any flaws permitted by DD-G-451 shall not interfere with reading of the dial.

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Angle A From Vertical Center Line	Ranges (PSI)		
	0 - 5	0-10	0-25
78°	0	0	0
142°	1	2	5
206°	2	4	10
270°	3	6	15
334°	4	8	20
38°	5	10	25

Ranges PSI		Angle A From Vertical Center Line
0-15	0-30	
0	0	78°.0°
	5	131.3°
5	10	184.7°
	15	238.0°
10	20	291.3°
	25	344.6°
15	30	38.0°

Ranges PSI	Angle A From Vertical Center Line
0-20	
0	75°
5	158°
10	238°
15	318°
20	38°

E-Z Mark at 238° All Indicators. All Dial Scales to be Linear Within ± 1/2°

Marking	Height or Length inch ± 0.016	Width of line or Graduation inch ± 0.005	Finish
Numerals	.12	0.020	Lusterless White
Major Graduations	.12	0.020	
Minor Graduations	.06	0.015	
Lettering "PSI"	.09	0.015	
Lettering Nomenclature	.12	0.020	
Background of Dial			Lusterless Black

Figure 1 - Dial

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3.7.1.1 Cover glass mounting - The distance from the inner surface of the cover glass to the surface of the dial on which the marking is applied shall not exceed 0.188 inch for single indicators or 0.250 inch for dual indicators.

3.8 Wiring - The electrical wiring system shall be insulated from the indicator case, except that "J" pin of the indicator shall be grounded to the case. The wiring, shall be as shown in Figure 2. (Pin "J" of the dual indicator is case ground and is intended to be connected to the airframe ground in service. Pins "A" and "K" may be externally jumped to pin "J" if it should become desirable to ground these circuits at the indicator. The mating connector should be used for this purpose).

3.8.1 Electrical connector - The electrical connector used shall conform to MIL-C-26482 or MIL-C-5015 and shall be of the type specified in MS17996. The connector shall be mounted on the rear surface of the case with the axis of the connector within 1/16 inch of the axis of the case. The largest polarizing keyway of the connector shall be at the top center of the indicator.

3.9 Identification of product - The indicator shall be marked for identification in accordance with MIL-STD-130 and shall include the following information:

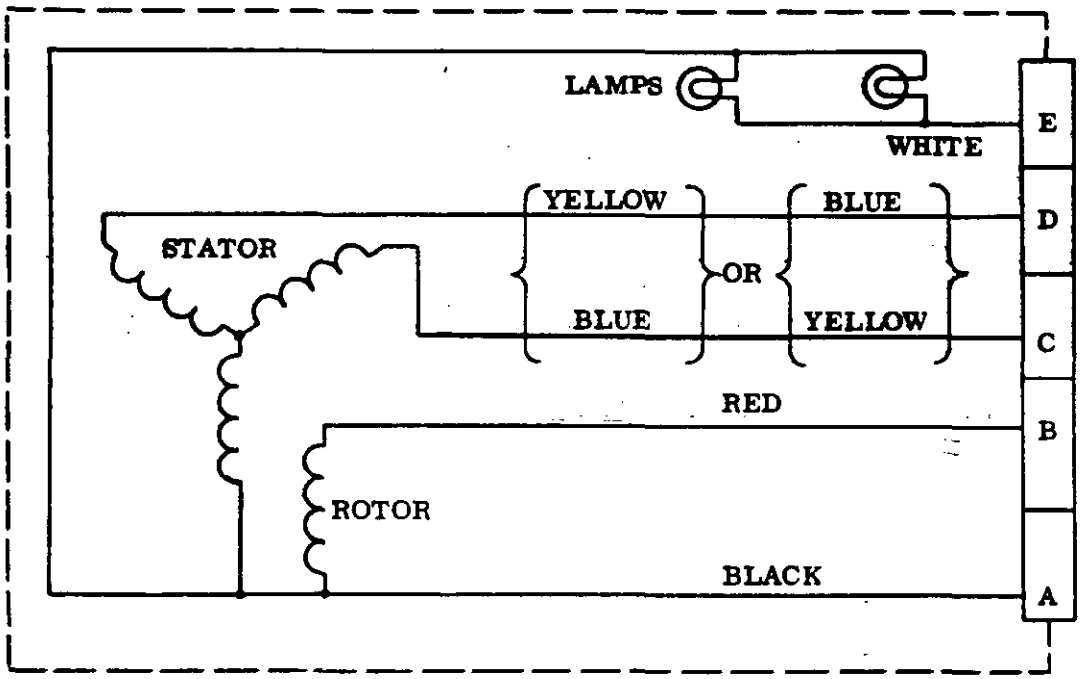
Indicator, Pressure, 1 1/2 - Inch, Integrally Lighted
MS part No.
Stock No.
Manufacturer's part No.
Manufacturer's serial No.
Contract or order No.
Manufacturer's name or trade-mark
US Property

3.10 Weight - The weight of the indicator shall not exceed 0.7 pound for the single indicator or 1.2 pounds for the dual.

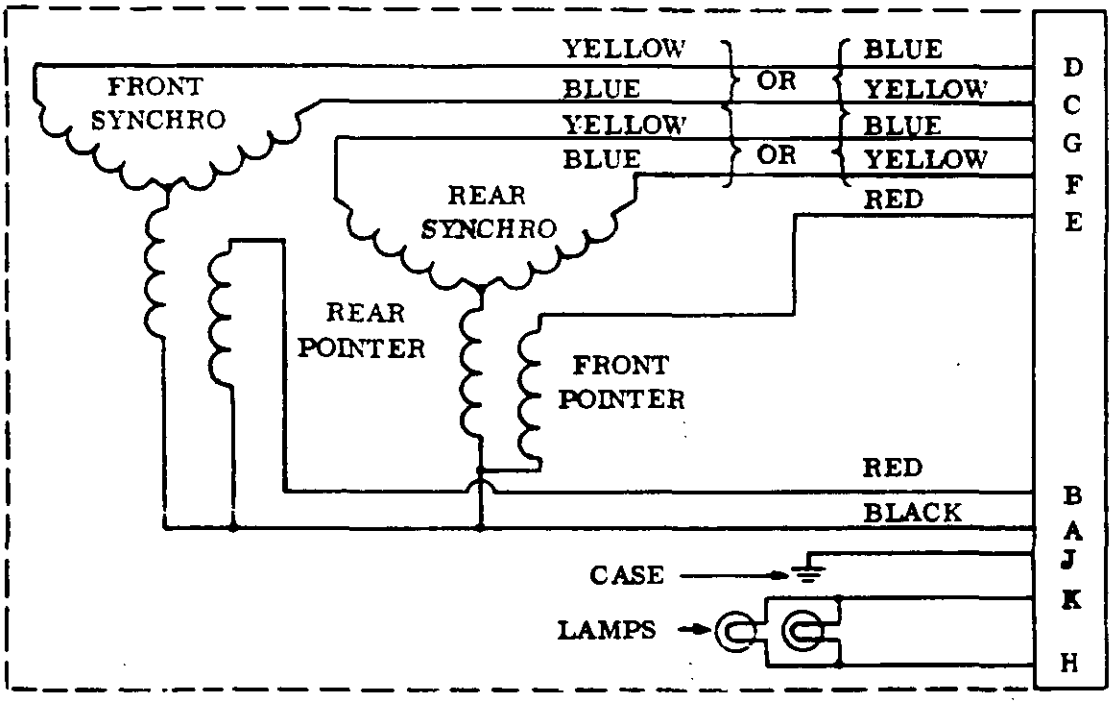
4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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SINGLE INDICATOR



DUAL INDICATOR

FIGURE 2 - INTERNAL WIRING

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4.2 Classification of inspection - Inspection of the Indicator shall be classified as follows:

(a) Qualification inspection: Qualification inspection consists of examinations and tests performed on sample indicators submitted for approval as a qualified product.

(b) Quality conformance inspection: Quality conformance inspection consists of examinations and tests performed on indicators manufactured and submitted for acceptance under contract.

4.3 Qualification inspection - The qualification inspection of the indicator shall consist of all the examinations and tests of this specification performed in the order specified under the paragraph headed Inspection methods and samples submitted for Qualification shall be accompanied by the suppliers report showing that the reliability test for the Reliability Qualification Phase (4.4.3) has been satisfactorily met on other samples of the same equipment.

4.3.1 Qualification inspection sample - Qualification inspection samples shall consist of three indicators manufactured in accordance with this specification. The indicators submitted for qualification inspection shall have been previously subjected only to the individual inspections. The samples shall be forwarded to the laboratory designated in the Letter of Authorization.

4.3.1.1 Qualification inspection sample identification - The qualification inspection samples shall be plainly identified by durable tags, securely attached, and marked with the following information:

Sample for Qualification Inspection
Indicator, Pressure, 1 1/2-Inch, Integrally Lighted
Submitted by (Manufacturer's name, date)
for Qualification Inspection in accordance with
Specification MIL-I-23479 under authorization
(reference letter authorizing tests).

4.4 Quality conformance inspection - The Quality conformance inspection shall consist of the Individual inspection, the Sampling plans, and the Reliability assurance tests of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing all the inspections. Sampling plan B inspections shall be conducted at a Government laboratory designated by the procuring activity. Quality conformance inspection, except for Sampling plan B, shall be under the supervision of the Government quality control representative. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

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4.4.1 Individual inspection - Each indicator submitted for acceptance shall be subjected to the Individual inspection. This inspection shall determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each indicator accepted shall have passed the Individual tests of MIL-I-7057.

4.4.2 Sampling plans - The Sampling plans shall consist of Sampling plan A and Sampling plan B inspections. The inspection samples selected for sampling tests shall first have passed the Individual inspections. The inspection samples which have been subjected to Sampling plan A inspection shall not be delivered on contract until they have been refurbished and resubmitted and passed all the individual inspections. Inspection samples which have been subjected to the Sampling plan B inspection shall not be delivered on contract.

4.4.2.1 Sampling plan A sample selection - Sampling plan A samples shall be selected at random in accordance with the following schedule:

Quantity Offered for Acceptance	Quantity to be Selected for Inspection
First 15	(See Note)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 of fraction thereof	1

NOTE: When Sampling plan B is invoked, the quantity shall be zero. When Sampling plan B is to be omitted, the quantity shall be one.

When a defective indicator occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of fifteen (when Sampling plan B is omitted) and proceed as indicated.

4.4.2.1.1 Sampling plan A inspection - Each sample selected for Sampling plan A inspection shall be subjected to the Sampling plan A tests of MIL-I-7057.

4.4.2.2 Sampling plan B Instructions - Three indicators shall be selected at random from the first fifteen (15) produced on contract and submitted within 10 days after manufacture. These indicators shall be forwarded, at the contractor's expense, to a Government Laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

Indicator, Pressure, 1-1/2-Inch Integrally Lighted
Submitted by (Manufacturer's name, date)
for production acceptance Sampling Plan B inspection,
in accordance with Contract or Order No.
Manufacturer's Part Number

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4.4.2.2.1 Sampling Plan B inspection - Each sample selected for Sampling Plan B inspection shall be subjected to the Sampling plan B tests of MIL-I-7057.

4.4.3 Reliability tests - The Reliability tests of MIL-I-7057 apply. Test Level F of MIL-STD-781 shall apply to both phases and Test Plan I to the Qualification Phase (Predelivery tests) and Test plan II to the Sampling Phase (Delivery tests).

4.4.4 Inspection conditions - The inspection conditions (Test conditions) of MIL-I-7057 shall apply.

4.4.5 Inspection methods - All inspection methods (Test methods) of MIL-I-7057 shall apply. Tolerances not given in MIL-I-7057 are given in Table I. The following are exceptions from the tests of MIL-I-7057.

Dielectric strength - Pin "J" of dual indicators shall be excluded in the procedure.

Magnetic effect - The tolerance shall be one degree in lieu of 3 degrees.

TABLE I

Pressure Indicator Tolerance PSI

Dial Scale	0-5	0-10	0-15	0-20	0-25	0-30
Scale error						
Room temperature	0.025	0.050	0.075	0.10	0.125	0.150
Low temperature High temperature	0.050	0.100	0.150	0.20	0.250	0.300
Friction						
Room temperature	0.025	0.050	0.075	0.10	0.125	0.150
Low temperature High temperature	0.050	0.100	0.150	0.20	0.250	0.300
Vibration						
Pointer oscillation	0.025	0.050	0.075	0.10	0.125	0.150
Pointer variation	0.025	0.050	0.075	0.10	0.125	0.150
Position Error at Mid-scale (EZ)	0.025	0.050	0.075	0.10	0.125	0.150

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A - Unless otherwise specified in the contract or order, Indicators, Pressure, 1 1/2 Inch, Integrally Lighted shall be individually preserved and packaged in accordance with MIL-P-116, Method III, in a unit container conforming to Type CF, Class Domestic, Variety SW, Grade 125, PPP-B-636.

5.1.2 Level C - Indicators, Pressure, 1 1/2 Inch, Integrally Lighted shall be individually preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. This level may conform to the supplier's standard practice provided the latter meets the requirements of this level.

5.2 Packing - Packing shall be Level A, B or C, as specified (see 6.2).

5.2.1 Level A - Indicators, Pressure, 1 1/2 Inch, Integrally Lighted preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. Shipments that exceed the weight limitations of this specification shall be packed in overseas type containers conforming to PPP-B-601. The gross weight of each container shall not exceed 200 pounds. Where practicable, exterior shipping containers shall be of uniform shape, size, and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of Specification PPP-B-636 or PPP-B-601, as applicable. Waterproofing shall be in accordance with MIL-STD-1186.

5.2.2 Level B - Indicators, Pressure, 1 1/2 Inch, Integrally Lighted preserved and packaged as specified in 5.1.1 shall be packed in domestic shipping containers conforming to PPP-B-636. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. The gross weight of each shipping container shall not exceed the weight limitations of the specification. Closure and strapping shall be in accordance with the appendix of Specification PPP-B-636.

5.2.3 Level C - Packages that require overpacking for acceptance by the carrier shall be placed in exterior type shipping containers in a manner that will insure safe transportation at lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules or Regulations of other carriers as applicable to the mode of transportation.

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5.3 Physical protection - Cushioning, blocking and bracing shall be in accordance with MIL-STD-1186, except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. Drop tests of MIL-STD-1186 shall be waived when preservation, packaging and packing of the item is for immediate use or when drop tests of MIL-P-116 are applicable.

5.4 Marking - Interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.4.1 Precautionary marking - In addition to any other marking required by order or contract, the following precautionary markings shall appear on two sides of each interior and exterior shipping containers:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended Use - The pressure indicators covered by this specification are intended for use on aircraft to provide remote indication of various pressures in the aircraft.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) The quantity, MS part number including dash number and type number, if any, of the instrument desired.

(c) Levels of packaging (see 5.1) and packing (see 5.2) desired.

(d) The laboratory that shall conduct tests (see 4.4, 4.4.2.2).

(e) Reliability test phases (see 4.3).

6.3 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for the opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D. C., 20360.

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6.4 International standardization - Units of measurement (3.4 and figure 1) of this specification are the subject of international standardization agreement ABC 10/2C. When an amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:

Air Force - 71
Army - AV
Navy - AS

Preparing Activity:

Navy - AS

Reviewer activities:

Air Force - 71
Army -
Navy - AS

Project No. 6620-0103

International interest (see 6.4)

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